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ERICSSON TECHNOLOGY Review

SPOTLIGHT ON

EXTENDED REALITY

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Time elapsed
Distance
Pace
Speed
148 bpm


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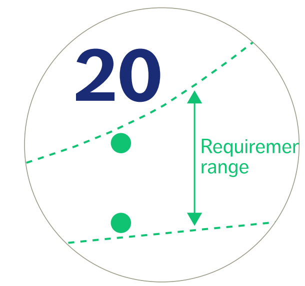
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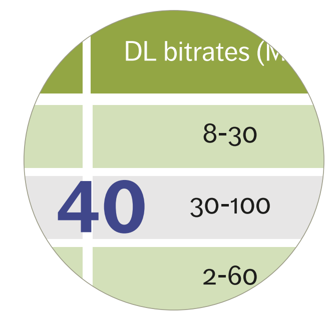
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Ericsson Technology Review brings you insights into some of the key emerging innovations that are shaping the future of ICT. Our aim is to encourage an open discussion about the potential, practicalities, and benefits of a wide range of technical developments, and provide insight into what the future has to offer.

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EXTENDED REALITY – A WORLD OF NEW CHALLENGES AND OPPORTUNITIES

■ **THERE IS INCREASING** consensus in our industry that extended reality (XR) – an umbrella term that covers immersive technologies ranging from virtual reality (VR) to augmented reality (AR) – could be the next paradigm shift after the smartphone. Since the network requirements of advanced XR use cases differ significantly from those of mobile broadband, there is every reason to believe that this shift will have a transformative impact on the communication industry. Ericsson is working closely with leading players around the world to shape the market and ensure that 3GPP (3rd Generation Partnership Project) technology is relevant in the XR ecosystem.

Many smartphone users have already experienced basic forms of AR by using the camera filters in apps like Snapchat and playing games like Pokémon GO. AR technology becomes much more powerful, however, when it is used with specialized devices such as AR glasses and other types of head-mounted displays (HMDs). AR HMDs transform the user interaction by freeing up the user's hands and making it easy to overlay information on top of the real world – capabilities that have been shown to increase worker efficiency dramatically. By making the internet available and integrated with the real world in near real-time, more advanced forms of XR will have the potential to deliver an even more immersive user experience.

> XR… COULD BE THE NEXT PARADIGM SHIFT AFTER THE SMARTPHONE

For simple AR applications that are already on the market today, such as digital overlays that help with street directions, current networks are sufficient. However, as XR applications become more advanced, the connectivity requirements will become more stringent. Introducing XR traffic in mobile networks puts high pressure on processing and transmission bitrates across the whole communication chain, which has an impact on the uplink, downlink and latency requirements. This shift will require high-performance networks with increased capacity, more uplink throughput and bounded latency. To support XR use cases, future networks will need to have the ability to handle a wide range of applications with varying streaming, spatial-mapping and edge computing requirements simultaneously.

We hope this special issue of our magazine helps you and your organization gain a better understanding of both the challenges and the opportunities that XR presents for your business. Please feel free to share it with your colleagues and partners. You can find both PDF and HTML versions of all the articles at: www.ericsson.com/ericsson-technology-review

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Future network requirements for extended reality applications

Many experts believe that new and emerging extended reality technologies will lead to the next major paradigm shift in telecommunications, with lightweight XR glasses ultimately overtaking smartphones as the dominant device type in mobile networks. This evolution has major implications on the requirements for future networks.

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The availability of discrete, attractive and high-performing extended reality (XR) devices is key to the mass-market success of XR applications, and the process of creating them is well underway. Device developers are working intensively to evolve the device form factor from today's bulky head-mounted displays (HMDs) to lightweight and stylish glasses that will depend on computation offloading to a much greater extent.

■ It is of the utmost importance that future networks are designed and dimensioned to meet the requirements of XR applications. Low latency, high reliability and high data rates will be essential to sustain user quality of experience (QoE) in XR applications while offloading computation services. XR applications are also going to generate a significant amount of additional traffic, which networks must have the capacity to handle.

To gain a better understanding of XR network requirements, Ericsson has derived a set of XR operating points as concrete samples from the requirements space, as illustrated in *Figure 1*. For the sake of simplicity, we refer to the selected samples (represented by the dots in Figure 1) as XR "flavors." We have used the flavors to make projections regarding how the network requirements of XR applications, devices and computation offloading variants are likely to evolve over time, and what implications the evolving requirements will have in terms of network capabilities.

To start, we identified which XR applications are most likely to be used and how advanced we expect them to be. In the next step, we considered computation offloading and its impact on connectivity. The XR flavors result from a combination of these two aspects, together with device capabilities, that enables us to derive the requirements on the network for these specific examples.

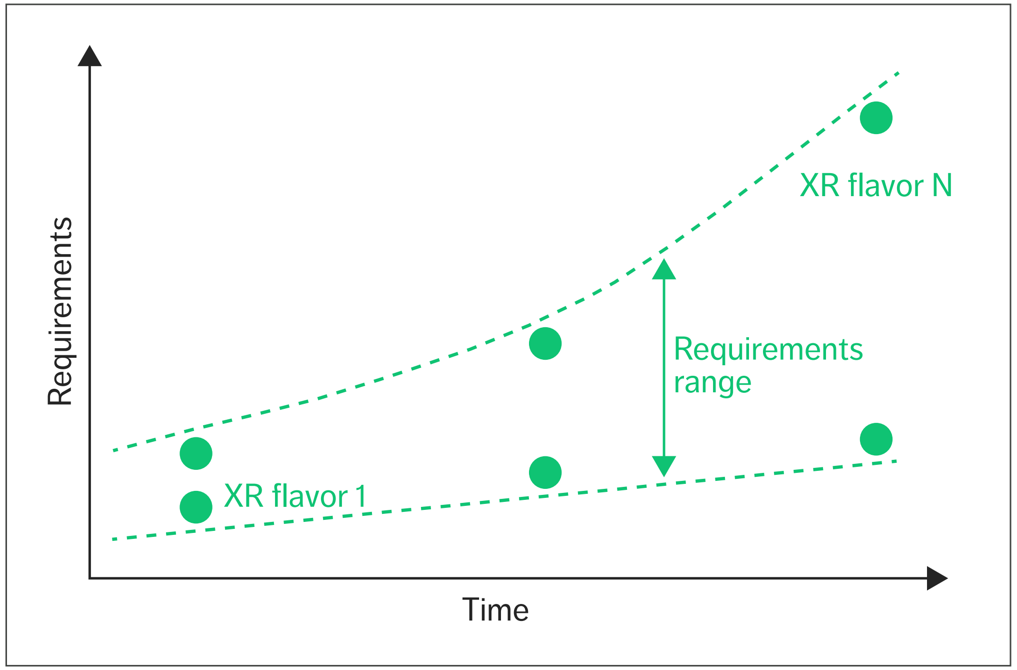

Figure 1 XR requirements with stringency increasing over time

XR applications and user behavior

Extended reality is an umbrella term that covers immersive technologies ranging from virtual reality

Terms and abbreviations

AR – Augmented Reality | **DL** – Downlink | **HMD** – Head-Mounted Display | **MBB** – Mobile Broadband | **Mbps** – Megabits per Second | **ms** – Milliseconds | **QoE** – Quality of Experience | **RAN** – Radio-Access Network | **SLAM** – Simultaneous Localization and Mapping | **UL** – Uplink | **XR** – Extended Reality

DAILY XR USAGE IS EXPECTED TO INCREASE SIGNIFICANTLY AS XR DEVICES BECOME SLIMMER AND MORE CONVENIENT TO WEAR FOR LONGER PERIODS

to mixed reality and augmented reality (AR) [1]. XR devices are expected to bring immersive experiences that merge digital objects, information and overlays with the real world. XR represents three major shifts:

1. from flat screens and 2D content to immersive 3D content displayed and spatially located in the real world
2. from graphical user interfaces to natural interaction within the real world
3. from context switching (between screen and real world) to context awareness.

Application areas that have a high potential for wide uptake include gaming and entertainment, retail and shopping, social communication and virtualized work. These comprise consumption of XR content in a single user setting as well as communication services such as holographic communication that include two or more XR users [2].

XR applications differ in terms of complexity of graphics, as well as on how aligned the graphics need to be to the real environment. The range of XR applications includes everything from a simple navigation service with an arrow pointing toward a destination to advanced holographic communication with moving 3D objects perfectly aligned with the real environment. Unsurprisingly, the most advanced applications with the highest degree of computation offloading have the greatest impact on network requirements, and the simplest ones with the least amount of computation offloading have the single application can be elastic in the sense that it can operate with different resolutions, for example.

Network requirements are also impacted by usage location and the distribution of use over the time of day. Consumer studies indicate that the introduction of XR may not lead to abrupt changes in user patterns – that is, consumers will likely continue to use the same type of services at the same locations and times as now. In this sense, XR is "only" a new type of user interface. Over time, XR is expected to create additional value for consumers across a broad range of application areas [3, 4].

Daily XR usage is expected to increase significantly as XR devices become slimmer and more convenient to wear for longer periods. But even when it becomes feasible for large numbers of people to wear XR devices all day, it is reasonable to expect the types of XR services that are used to vary over the course of the day, similar to the usage patterns of smartphone services today. Therefore, we expect it would be adequate to dimension future networks to support heavy XR usage for as little as one hour, or just a few hours, per day.

XR functions and computation offloading

Many of the functions needed to realize XR experiences are computationally heavy. Rendering and spatial computation are the two most important. Rendering refers to the process of generating graphics to be shown to the user in the HMD. Depending on the complexity and resolution of the graphics, rendering can become very computationally demanding. Spatial computation functionalities include simultaneous localization and mapping (SLAM), object detection and object tracking. These are primarily used to gain an understanding of the local environment.

Rendering and spatial computation are both good candidates for offloading to a remote server – that is, an edge or cloud server with adequate computation capability. Computation offloading enables more powerful processing through cloud computation and reduced HMD complexity in terms of a smaller form factor, increased battery life and reduced heat

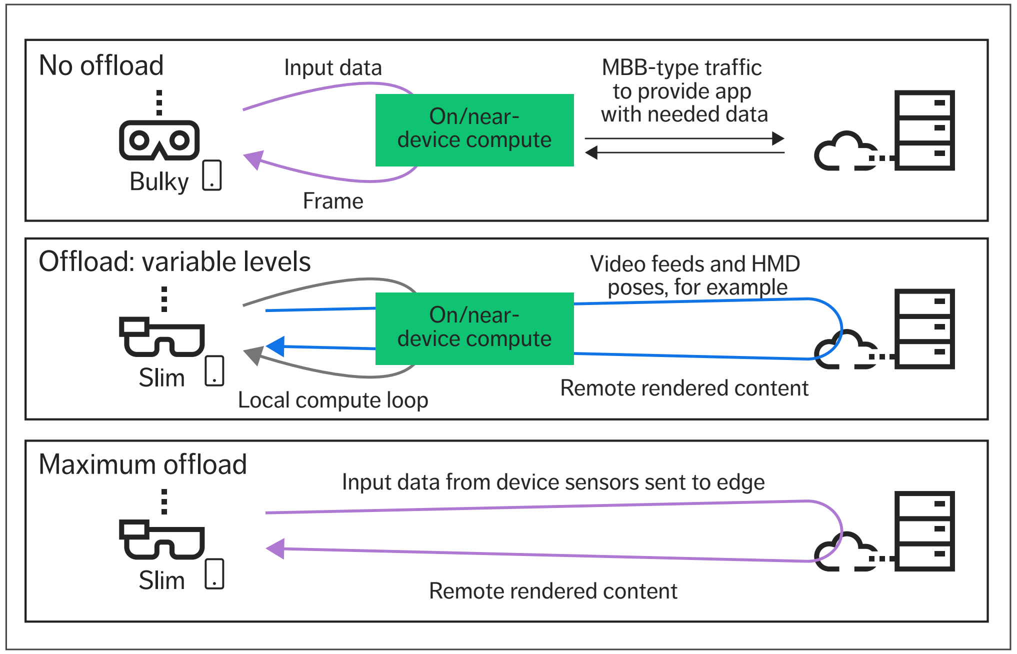

Figure 2 Three approaches to XR computation processing

complex (and faster) computations that enable more immersive experiences with higher resolution, quality and frame rates, as well as supporting multi-user experiences in dynamic environments.

Figure 2 illustrates the three main approaches to XR computation processing. In the "no offload" scenario at the top, all the computation occurs in the device itself or with the support of a tethered companion device nearby (such as a smartphone). In the "maximum offload" scenario at the bottom, all computation occurs remotely, in the cloud. Previously referred to as a "high offload" option [1], maximum offload has extreme requirements on connectivity and is not expected to reach mainstream adoption in the foreseeable future.

In the "offload: variable levels" scenario in the middle, some computation occurs in or near the device, while other computation occurs in the cloud. This split may be user-specific and possibly dynamic capabilities, cloud capabilities, network functionalities and conditions, and user preferences. This scenario corresponds to the "low offload" and "mid offload" options defined in a previous article [1] and represents the set of offload scenarios we expect will gain market traction.

Offloading as a mechanism implies data transmission between device and the processing unit on the network side, where the resulting bit rate and latency requirements vary depending on many factors.

In the case of remote rendering, the application graphics are rendered at a remote server (an edge server, for example) and sent to the HMD through a 5G (or other) network. Cloud gaming is a contemporary example of this. Remote rendering requires high downlink (DL) bitrate with bounded low delay to provide good QoE. Typically, two or more 2D video streams are sent in the DL. When different resolutions will be sent to the HMD. Another version of remote rendering is to render volumetric content remotely and let the HMD make the final 2D rendering based on the most up-to-date pose data. In this case, HMD-local, latency-hiding techniques such as space warp may use the 3D data streams to enable the user to look at an object from other angles without requiring more data in the DL.

When offloading spatial computation functions, the HMD uploads compressed sensor data (video, lidar and so on) or detected objects to a server that builds a map of the environment and positions the HMD within it. Offloading SLAM enables shared immersive multi-user experiences, as the map of the environment can easily be shared between users in the remote server. These maps can make the positioning task of an HMD that is entering a previously mapped environment more efficient and allow for object persistence. There is a variety of offloading options for SLAM, and complete offload implies very stringent latency requirements. The uplink (UL) bitrate depends on the resolution and level of compression of the sensor data (or feature data) and can be large.

For object detection and tracking, a video feed is typically uploaded to a server for image recognition and tracking of previously identified objects. The location of the objects, and possibly other metadata, is sent to the HMD. Object tracking places stringent requirements on the latency if the function is offloaded. Offloading only object detection could be less sensitive to latency, as the corresponding information can be sent to the HMD when it becomes available. Then the latency-sensitive object-tracking functionality on the HMD takes over to ensure that the visual information presented to the user stays aligned with the real world.

In cases where processing is done in the device and/or in a tethered companion device, the communication over the network is limited to application data, with characteristics that are similar to mobile broadband (MBB) traffic. The real-time processing power is mainly limited to what is available in the glasses and the companion device. The device capabilities are much more limited when running without a companion device with local processing only in the HMD. The processing in slim-form-factor AR glasses is expected to only be sufficient for basic SLAM and rendering, while more of the SLAM needs to be performed remotely for immersive and interactive experiences. For the cases with complete offload of all XR application processing functionality, the HMD will always have to run some processing locally, including overlay imaging, localization and latency-hiding processing.

In short, smaller XR device form factors, longer battery life and advanced XR services will require more cloud computing due to the increased need for computation offloading. The impact on network requirements will vary depending on which XR functions and/or their various functionalities need to be offloaded. It makes sense to offload XR functions that enable the highest device power savings or the greatest QoE improvements, as long as the connectivity requirement in terms of bitrates and latencies can be met by the available networks. Offloading will then be progressive as networks evolve and become more capable to meet the various

Key terms

Foveated rendering is a technique that uses eye-tracking to identify where a user is looking and render higher resolution images in those directions, while rendering lower resolution images for peripheral vision. The purpose is to save bandwidth and reduce the rendering complexity.

Object persistence means that a virtual object remains in the same location, even if the user moves out of the environment. A placed object could be stored in a server along with a precise position in a stored map. The map can be tagged with its location in the real world. The storing of detailed maps of scanned environments comes with privacy concerns and is a topic of ongoing research and regulation.

XR NETWORK REQUIREMENTS ✱

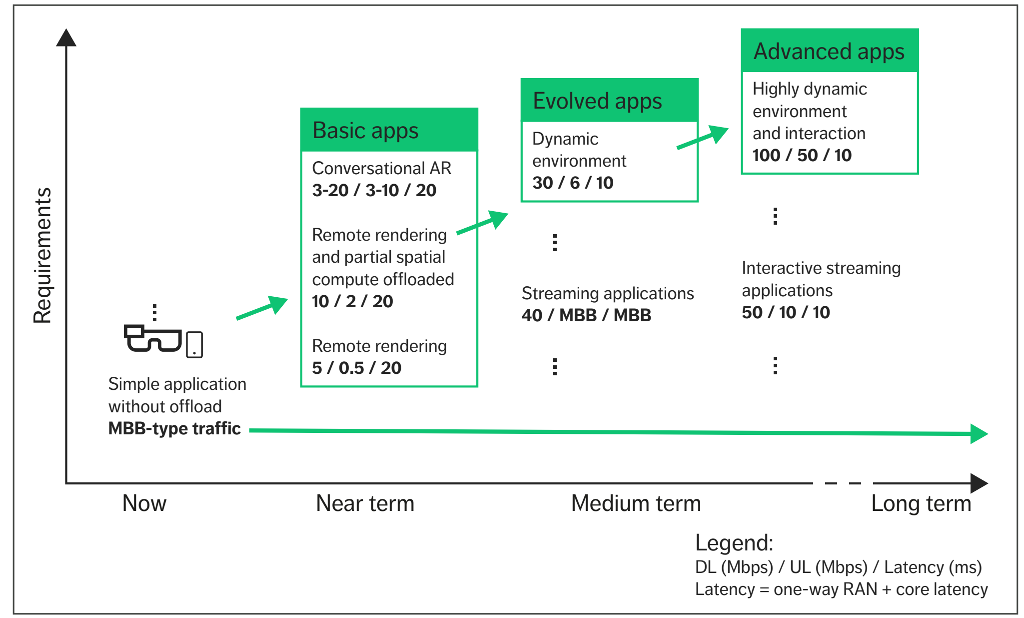

Figure 3 Evolution of requirements over time

and changing requirements of XR services and offloading levels.

Example operating points – XR flavors

A large multi-dimensional space results from the combination of a broad range of more and less advanced XR applications and various XR device capabilities that are associated with different levels of computation offloading. In theory, any point in this space is a possible XR flavor with its own requirements for network support. To simplify the analysis of required network solutions we narrow our focus to a smaller sample. The sample of XR flavors that we present here is based on concrete experiments with various partners [5, 6] and our own research and has been chosen to represent a potential evolution of slim AR glasses. The time dependency of the dimensions is a key consideration.

The XR flavors that we have chosen to analyze should be understood as a limited selection of samples in a wide range. They are examples rather than forecasts, and the conclusions we draw cannot be directly extrapolated to all XR applications and all cases. The requirements that we have obtained from the different XR flavors are simplified and intended to provide general guidance. In reality, XR requirements will be very elastic in both DL and UL bitrates, as well as in latency, at least to some degree. Thus, it is expected that XR requirements will be defined by a range of values as opposed to a single value.

In *Figure 3* we present the XR flavors along a time axis and a strictness-of-requirements axis. For each point in time – now, near term, medium term and long term – there are flavors with different characteristics, resulting in different requirements on the network. Increasingly advanced XR applications will become available over time, which will, together with the device capabilities, place increasingly stringent requirements on the connectivity. Still, at each time instance, a wide range of applications will be used, some of which will be very simple, while others will be more complex. The

> THE CONVERSATIONAL AR FLAVOR THAT WE HAVE CONSIDERED IN THE NEAR TERM INVOLVES NO COMPUTATION OFFLOADING

bitrate requirements shown in Figure 3 are aggregated over the flows, whereas the one-way delay requirements are for the most sensitive flow.

Now – simple XR applications without offload

In the simplest XR applications, for which offload provides limited benefit, all processing is done locally on the HMD or on a tethered device. This is similar to how AR applications work on mobile phones today. The requirements are much like those of MBB, and there are no stringent requirements on latency. Example applications include simple navigation services and the ability to extend the touch-screen user interface of smartphones and tablets beyond the bounds of their physical screens with virtual screens controlled by hand gestures. Relevant parts of the global representation of the world are downloaded without real-time requirements. The global map used by these applications is updated on a relatively slow timescale. The time lapse before a newly-created AR object – a new restaurant in a navigation app or a user-added static virtual object in an AR geocaching app, for example – becomes visible for other users depends on how often the global map is updated based on user/actor input and how often the local map is refreshed from the global map.

Near term – basic XR apps

For simple XR applications with remote rendering, we assume that SLAM is run on the HMD. The remote-rendered content would be 2D video content or very simple volumetric content sent in the DL to the HMD. The DL bitrates vary depending on the resolution, frame rate and level of compression. The UL traffic consists of the HMD pose and user input, which is sent to the remote server to ensure proper rendering. This category of devices would require a relatively low UL rate of 0.5Mbps. A slightly higher DL rate would be needed depending on the size and resolution of the virtual content created in the edge cloud. We chose a sample point of approximately 5Mbps. The one-way latency for such a use case would be estimated at around 20ms, including the radio-access network (RAN), the core and the transport network. In this flavor, exchanging anchor points between users enables the alignment of coordinate systems and local multi-user applications such as board games.

In simple XR applications that offload both rendering and partial spatial computation, localization occurs in the HMD, while spatial mapping, map optimization and object detection are offloaded. This means that 3D objects are rendered on the remote server, and 2D images for the displays are rendered in the HMD. This setup provides the HMD with 3D content and allows for local adjustments in the rendering to account for the user's head movements as late as possible in the rendering chain. The DL throughput depends on the compression and resolution of the 3D content. We use 10Mbps as an estimate of what could be realistic for low-resolution 3D content.

The UL data is composed of the HMD pose, user input and data representing the detected environment in the form of UL video and sensor data, for example. The UL bitrate depends on the resolution and frame rate of the video and sensor data. A more detailed representation of the detected environment requires better maps and more precise localization. Parts of the spatial mapping and remote rendering have an impact on latency requirements. The offloaded spatial computation functions of this flavor enable an increased understanding of the local environment.

The conversational AR flavor that we have considered in the near term involves no computation offloading. The requirements focus on sending a volumetric data stream generated by one device to receiving HMD(s). At one end, a person is continuously scanned and a volumetric data stream is generated and sent to the receiving HMD(s) for display. In this scenario, we assume that the rendering from volumetric data to the HMD is performed on the HMD. The UL and DL bitrates depend on the resolution of the point cloud, the frame rate and the level of compression. A reasonable operating range in the near term would be up to 20Mbps in the DL and 10Mbps in the UL. The DL bitrate will depend on the number of participants in the conversation.

Medium term – evolved XR apps

The requirements to support more advanced applications will be more challenging, particularly in dynamic environments where AR objects must adapt to changes in the physical world in real time. One example would be to have a virtual AR character, such as a Pokémon, walk alongside a user on a sidewalk and avoid colliding with anyone or anything around it. Both remote rendering and the offload of spatial mapping will likely be required to achieve this. It will be essential to update the digital representation of the physical environment quickly enough for the application engine to respond with a timely and appropriate rendering of the AR object. The latency requirements in both the UL and DL for this flavor are much tighter than for previous ones, because the consequences of failing to detect and map certain parts of the environment with sufficient quality are greater. For example, the AR experience would be seriously undermined if an object in front of the Pokémon were left undetected and the Pokémon walked through it rather than jumping over it.

To avoid such error events, the representation of the detected environment needs to increase in size (in terms of higher sensor resolution and higher sensor frame rates, for example), which leads to an increase in the UL bitrate compared with previous flavors, based on the assumption that the content sent in the DL is remotely rendered, high-quality 2D video streams.

In the medium term, we also expect to see streaming applications evolve to use XR. For example, content-providing apps such as Netflix, TikTok and Snapchat may provide volumetric video streams that offer more immersive 3D-viewing experiences such as the Star Wars movie scene where Princess Leia is projected by R2D2 as a hologram. In this flavor, SLAM is handled on the HMD, along with decoding to 2D. The DL requirements would vary depending on the video quality and frame rate. A point cloud stream of 40Mbps could represent two digital objects at a given resolution, for example. The UL bitrate and latency requirements for this flavor are similar to MBB traffic requirements.

Long term – advanced XR apps

In the long term, we expect that interactive streaming applications will have the ability to both stream volumetric video and enable user interactivity – that is, the user's behavior will affect the content that is streamed from a server. In this flavor, we assume that SLAM will be offloaded. Rendering of the volumetric video stream, based on user interactions, will be performed at a remote server. The DL bitrate will depend on the complexity, resolution and frame rate of the content. The UL will be dominated by the HMD sensor data that is needed to offload SLAM. Low latency will be required both for SLAM offload and for user interactions with the content.

In future advanced XR apps that support highly dynamic environments and interactions, AR objects will be expected to adapt to the dynamic environment, moving as if they were physically present in it. It will also be possible for multiple users to perceive and interact with the AR objects simultaneously. The AR object casts shadows, is occluded by physical objects and is shown correctly

> THE AR OBJECT CASTS SHADOWS, IS OCCLUDED BY PHYSICAL OBJECTS AND IS SHOWN CORRECTLY AS A NATURAL PART OF THE PHYSICAL WORLD

as a natural part of the physical world. In this flavor, we assume a dynamic multi-user application with several users in the same environment that interact with each other and the AR objects. The digital world is shared in the cloud.

This XR flavor will require the streaming of high-quality volumetric content in the DL, which could become very large. Alternatively, high resolution and high frame rate remote-rendered 2D video streams could be considered, with or without foveated rendering. To enable correct rendering and spatial understanding, large sets of high-resolution HMD sensor data would be sent in the UL to the cloud. Spatial mapping would be offloaded and correspond to roughly 10-20Mbps in the UL. In addition, high-resolution 3D point-cloud data would be sent in the UL to enable an enhanced experience. This could correspond to 30-40Mbps. Latency requirements would be as strict as those of the dynamic environments flavor in the medium term.

Conclusion

The use of extended reality (XR) technologies is expected to rise significantly in the years ahead, generating new network requirements with varying degrees of stringency depending on device capabilities, how much computation offloading is required, and which specific applications will be used. While these future requirements are heterogeneous and uncertain, they are essential inputs to the process of developing new network capabilities to support future XR use cases. To assist us in this work, we have defined a set of XR flavors to serve as samples in the expected requirements range. Our detailed analysis of these flavors demonstrates that they will require networks that can support bitrates of tens of Mbps together with latencies of 10-20ms to be achieved with high reliability – a task that is significantly more difficult than that of supporting today's mobile broadband services.

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Further reading

- **Ericsson blog, Welcome to XR immersive experiences, available at:** *https://www.ericsson.com/en/blog/2021/11/welcome-to-xr-immersive-experiences*
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Network evolution to support extended reality applications

Networks must evolve to handle the traffic generated by extended reality applications. Beyond volume, this new traffic has some important differences compared with mobile broadband traffic. In particular, the need for low latency and high reliability combined with high data rates challenges the capabilities of current mobile networks.

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Enhanced capabilities in terms of both functionality and deployment will be essential for mobile networks to provide a good user experience for extended reality (XR) applications while continuing to deliver high-quality mobile broadband (MBB) and voice services.

■ As part of the process of defining the new capabilities needed to meet the network requirements of emerging XR applications, Ericsson has defined a set of combinations of XR applications, compute offload levels and device capabilities that we refer to as "XR flavors." [1] Together with projections on XR uptake, the XR flavors form the basis for requirements on network connectivity. The flavors are represented by the dots in *Figure 1*, which shows them in the context of both increasing XR requirements over time and the new network capabilities that will be necessary to meet those requirements.

The specific traffic characteristics of XR, together with their effect on coverage and capacity, are essential inputs to defining the different kinds of network functionality needed to support XR traffic, live up to XR requirements and meet the expected performance.

In addition to knowledge about the traffic generated per active XR user and the associated performance requirements, predictions about XR uptake among users and daily usage are also necessary to accurately assess network impacts. These two aspects are difficult to predict, but based on the existing research [2], we assume an uptake of up to 6 percent over the next couple of years. For our

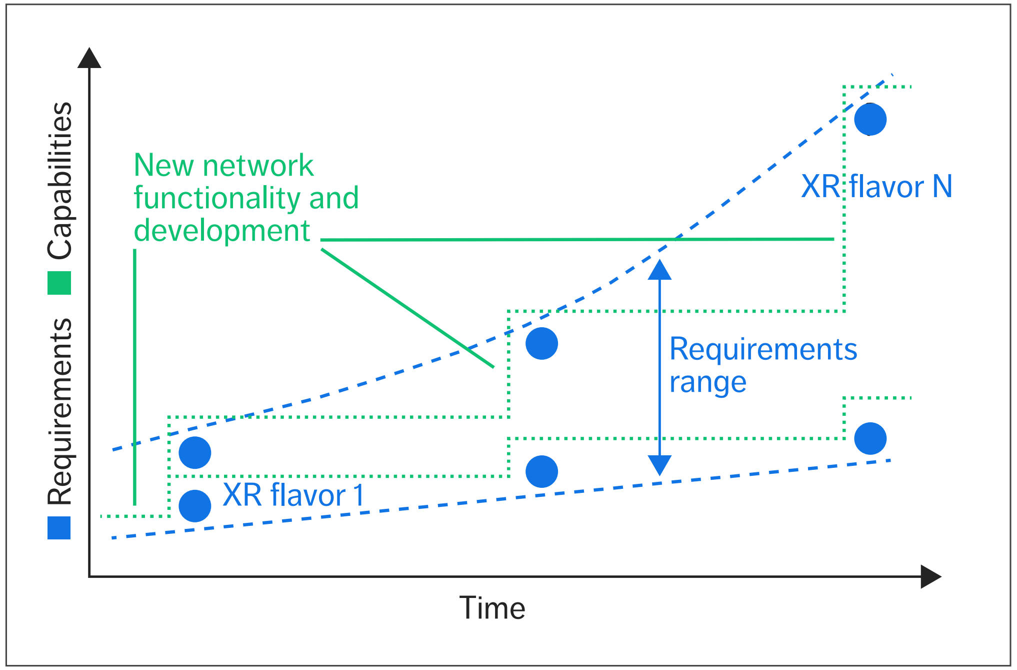

Figure 1 XR flavors in the context of both network requirements and capabilities over time

Terms and abbreviations

3GPP – 3rd Generation Partnership Project | **BSR** – Buffer Status Report | **DL** – Downlink | **DRX** – Discontinuous Reception | **FDD** – Frequency Division Duplex | **ISD** – Inter-Site Distance | **MBB** – Mobile Broadband | **Mbps** – Megabits per Second | **ms** – Millisecond | **QoS** – Quality of Service | **RAN** – Radio-Access Network | **TCC** – Time-Critical Communication | **UE** – User Equipment | **UL** – Uplink | **XR** – Extended Reality

NETWORK EVOLUTION FOR XR ✱

XR TRAFFIC IS CHARACTERIZED BY MULTIPLE SERVICE FLOWS, ARRIVAL JITTER AND A WIDE VARIATION IN PACKET SIZES

evaluations, we have chosen to use a near-term uptake of 2 percent and a mid-term uptake of 5 percent. With regard to the daily usage, we assume that during the hours of the highest traffic, the usage of the most demanding XR variants (which we use to dimension the network) approaches 10 percent [1]. Active XR users generate traffic according to the bitrates of the XR flavor.

XR traffic characteristics present new challenges

XR traffic is characterized by multiple service flows, arrival jitter and a wide variation in packet sizes [3]. When combined with stringent requirements on bitrates, delay and reliability, these characteristics have a big impact on a network's ability to support XR services.

XR traffic constitutes several different service flows, including video, audio, pose and control. Different XR applications may use a different number of flows, each with its own traffic characteristics and requirements. For example, pose may have a 10ms latency requirement, while video could have 20ms or 30ms. For good efficiency, the network should adapt to the needs of the individual flows. Today, multiple flows are commonly multiplexed into a single UDP stream, a practice that will need to be revisited to enable better network efficiency.

Arrival jitter is an issue for network support of XR because XR video traffic is typically quasiperiodic, which means that packets arrive at the network with slightly irregular small offsets from the average interval of arrivals. This is caused by delay variations in application encoding and transport networks [4]. The arrival jitter complicates the delivery of the packets within the bounded latency, as it consumes parts of the delay budget and makes precise allocation of resources in advance more difficult. Arrival jitter also affects several other aspects such as the discontinuous reception (DRX) configuration used to reduce device battery consumption, measurements and scheduling decisions.

The wide variation in packet sizes for XR traffic further complicates the allocation of resources. Depending on the flavor, a substantial part of XR traffic both in the downlink (DL) and the uplink (UL) is video, which is commonly characterized by a mix of packet sizes. Typically, new scenes or larger changes to a scene result in large packets, followed by a series of smaller packets reflecting smaller changes to the scene. In order to handle the largest frames, the network needs to be dimensioned for the peak bitrate. As changes of the scene are not known to the network, frame sizes appear random to the network, which complicates resource allocation.

In short, the combination of the different types of flows, the large packet-size variance even within the same flows, and the arrival jitter, together with the latency and reliability requirement, presents a significant challenge for today's mobile networks, which are designed for the much more relaxed delay budgets of MBB traffic.

Time-critical communication features for XR traffic

With each new generation, mobile networks have evolved and become more efficient in supporting MBB services. To maintain that efficiency while also ensuring that XR applications work well in 5G networks, it will be essential to separate the XR traffic from best-effort MBB traffic using the 5G Quality of Service (QoS) framework with optimized QoS flows [5].

The 5G QoS framework enables the use of the 5G time-critical communication (TCC) toolbox, which consists of both standardized and non-standardized functionality such as Low Latency, Low Loss, Scalable (L4S) throughput technology [6], UL configured grants, robust link adaptation and tailored scheduling schemes. These tools enable the optimized treatment of XR throughout the mobile network and specifically in the radio-access network (RAN) to mitigate the different sources of delay. Note that different tools may be applied to different flows.

Even with the help of these tools, however, we can expect lower capacity and worse coverage for some XR flows – and for low-latency, high-reliability services in general – compared with MBB. There are a few reasons for this.

Firstly, fading and interference variations cause the channel quality to vary over time in a way that is difficult to predict. Errors occur when we overestimate the channel quality and select an overly aggressive modulation and coding scheme. In the case of MBB, it is possible to use the more generous delay budget to achieve high efficiency nonetheless. We can wait for instances of good channel quality, use aggressive modulation and coding, and repair errors with multiple retransmissions, if needed.

The situation for XR is different. To support reliable low latency for XR, we need to use more robust link adaptation with an increased safety margin to avoid an overly high frequency of errors on the radio interface and reduce the number of retransmissions. Retransmissions are still essential to efficiently support high reliability, but each retransmission has a delay cost in terms of the round-trip time for sending and responding to a negative acknowledgement. This means that the number of retransmissions required to successfully deliver data needs to be reduced compared with MBB to meet the stricter delay bound. The tendency to underestimate the channel quality directly impacts coverage and capacity. Because of this impact, the level of robustness of the link adaptation should be set according to the requirements and not be overly conservative.

Lower capacity and worse coverage for XR are also caused by protocol aspects. It is not possible to use the whole delay budget to send data, since channel access and retransmissions also need to be accounted for within the delay budget. This means that the rate of data transmission over the air interface must be higher than the data transmission rate of the application. Due to the arrival jitter and unknown packet size, the data transmission needs to be preceded by a scheduling request and a buffer status report (BSR), in which the user equipment (UE) tells the network that it has something to transmit and how much data it wants to transmit. This takes time, which means that data transmission over the network cannot start the instant the data arrives from the application. The problem can be mitigated by pre-allocating resources by means of configured grants or smart dynamic prescheduling, which the device will use to transmit the BSR. Channel access latency can be reduced, as the UE can skip the scheduling request procedure, leaving more time for data transmissions.

BSR resource allocation accuracy and UE queueing delays are more critical to the capacity of XR service in comparison with MBB. Large and variable packet sizes may cause the UE to report high buffer status indexes that are inaccurate. In these situations the network overshoots the grant size, which often results in resource wastage. The lower the BSR accuracy, the larger the waste, and vice versa. An increased UE-internal queueing delay results when packets with no chance of meeting the latency requirements remain in the buffer for transmission, even if they are not useful for the application, worsening the delays for other packets that could still meet their latency requirements.

Improving standard support for XR

While much of the functionality described in the previous section can be implemented based on current 3rd Generation Partnership Project (3GPP) standards, some would benefit from enhanced standard support. To build an industry consensus, 3GPP has recognized a set of XR requirements and traffic characteristics [4] similar to ours and is using it as the basis for developing improvements. These improvements include XR awareness in RAN, and capacity and power saving enhancements that consider support for high data rates in combination with bounded latency requirements. The XR awareness in RAN is intended to make the RAN aware of both the XR traffic characteristics and the

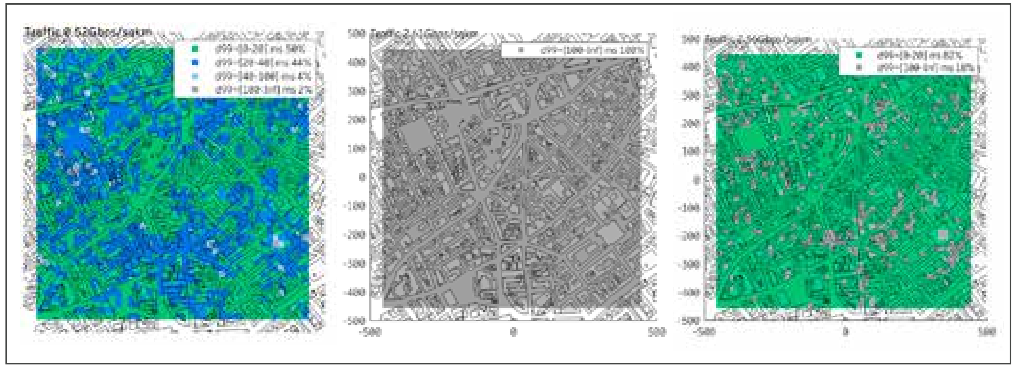

Figure 2 User satisfaction measured as DL packet delay fulfilled with 99 percent reliability

requirements of the XR application, as well as any changes to either of them in terms of the number and types of flows, minimum/maximum/average packet size for each of the flows, flow periodicity, jitter, and the association between IP packets and application packets, for example.

Capacity and power-saving enhancements are also being standardized in 3GPP Release 18. These are grounded in XR awareness in the RAN, which will likely continue to be the baseline for future RAN features specified in 3GPP. The capacity enhancements in 3GPP Release 18 address delay and resource wastage issues by including more precise buffer status reporting with delay information, supporting the dropping of application packets with no chance of meeting delay requirements and studies of dynamic configured grant adaptation.

The power-saving enhancements include adaptations to XR traffic characteristics of DRX, allowing a better matching of monitoring and sleeping periods to the XR traffic characteristics. We expect studying of further enhancements in future releases of the standard.

XR network simulations

To exemplify the ways in which networks will need to evolve to efficiently support XR traffic, we have run several simulations that demonstrate the performance of a range of XR flavors in different networks with varying characteristics in terms of site density and the type of environment.

Methodology

Our methodology is to first expose a network, as it is, without special functionality or network densification, to XR traffic and measure the XR user experience. In the second step, we introduce TCC functionality to determine the extent to which it can improve the user experience. Then we repeat the procedure for all networks and all the XR flavors [1].

Assumptions

The subscriber density in our simulations is 10,000 subscribers per square kilometer (1,000 for suburban) and 80 percent of them are indoors. Subscribers generate MBB and XR traffic according to the characteristics of the XR flavors and the assumptions on uptake and daily usage. We assume that 70 percent of the traffic is carried by in-building systems such as the Radio Dot System or Wi-Fi. Macro base stations are deployed with different densities in the different networks.

Average inter-site distances (ISDs) range from 150m to 1,000m. 3GPP New Radio is used on low and mid band, with frequency allocations of 2x10MHz frequency division duplex (FDD) at 800MHz, 2x20MHz FDD at 1,800MHz and 2,600MHz, and 1x100MHz time division duplex at 3,500MHz. Massive MIMO (multiple-input,

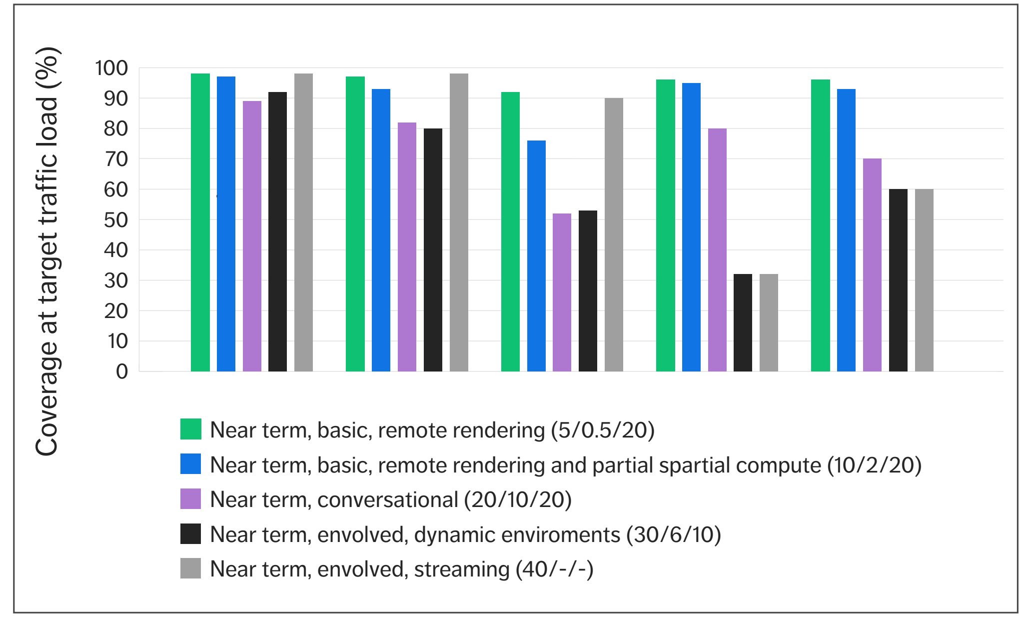

Figure 3 Coverage at target traffic load for five different XR flavors on five different networks

multiple-output) is used on 3,500MHz and regular sector antennas on the other bands. The TCC functionality used is related to service differentiation, admission control, delay-aware scheduling, robust link adaptation and configured grants for BSR. The same functionality is applied to all flows.

Results

Figure 2 illustrates the results for the XR flavor "near-term, basic with remote rendering and partial spatial compute" [1] in a macro network typical for many urban low-rise areas. We present the results as user positions in a three-dimensional map covering one square kilometer, colored by an estimated user experience, where green is good, gray is bad and blue is in between. The user experience is measured as the packet delay that can be supported with 99 percent reliability. A green marker corresponds to a position where 99 percent of the packets are delivered within 20ms, which for this flavor is assumed to lead to an acceptable user experience.

In the case on the left of Figure 2, traffic load is low and no TCC functionality is used. While there are several green markers, the coverage is far from full. In the middle of Figure 2, the traffic load increases to a level that we expect to see in the near future. This causes the map to turn entirely gray, indicating excessive delays, which we attribute to XR traffic queuing behind MBB traffic.

The right side of Figure 2 shows what happens when TCC functionality is added. The map turns predominantly green, with some gray areas that correspond to users who are not served either due to poor coverage or XR overload in those areas. In this scenario, the next step in the network evolution would be the use of capacity and coverage enhancing features on existing sites, or dedicated deployment in those areas.

Figure 3 presents a summary of the results we achieved when we repeated the analysis for five different XR flavors on five different networks that we refer to as Asian high-rise, US mid-rise,

DELAY-AWARE SCHEDULING AND ROBUST LINK ADAPTATION ARE EFFECTIVE TECHNIQUES TO MEET XR REQUIREMENTS

European mid-rise, suburban and 3GPP urban macro. The numbers in brackets represent DL bitrate in megabits per second (Mbps), UL bitrate in Mbps and one-way delay in milliseconds (ms). Coverage is measured as the fraction of users fulfilling the quality requirements in both the DL and UL at the projected traffic load. Three different points in time are assumed, and for each point in time a basic XR flavor and a more advanced XR flavor are covered.

In all five networks, the most basic XR flavor achieves good coverage of between 92 and 99 percent. For other flavors, the coverage varies significantly between the networks. In the Asian high-rise scenario, the network with the highest site density, most of the flavors achieve relatively good coverage, including the more demanding ones. The US mid-rise network is not far behind. Coverage drops for the more sparsely deployed networks, especially for the flavors with higher UL bitrate requirements or lower delay requirements. In these cases, an upgrade of existing sites and a complementary deployment in areas of outage would improve coverage. The impact of densification is roughly indicated by comparing the performance between networks with different site density (represented by the ISD).

Conclusion

The ability to support extended reality (XR) technologies in mobile networks will depend on the availability of new network capabilities that can meet stringent XR requirements on latency, reliability and bitrates. Our research indicates that delay-aware scheduling and robust link adaptation are effective techniques to meet XR requirements in the near term. Newly standardized features will also assist in meeting the XR requirements.

Our network simulations indicate that many modern networks are already able to support basic XR flavors in terms of coverage and capacity, and that the most densely deployed networks are also likely to be able to support more advanced flavors. For less densely deployed networks, the ability to support advanced flavors will likely require the upgrading of existing sites with enhanced coverage and capacity features, along with targeted densification in the case of the most sparsely deployed networks.

As network densification is complex and time-consuming, XR application developers who want full coverage should be encouraged to develop applications that are designed to adapt to varying network conditions.

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Further reading

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NETWORK EVOLUTION FOR XR ✱

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Holographic communication in 5G networks

Once confined to the realm of science fiction, holographic communication now ranks as one of the most wanted 5G-enabled applications by both consumers and enterprise users, according to recent research.

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Holographic communication refers to real-time capturing, encoding, transporting and rendering of 3D representations, anchored in space, of remote persons shown as stereoscopic images or as 3D video in extended reality (XR) headsets that deliver a visual effect similar to a hologram.

■ After several years of experience with video calls on smartphones and tablets, many users report that they are eagerly awaiting the chance to meet others digitally using immersive communication services such as 3D holographic augmented reality (AR) calls over 5G [1]. Compared with flat screen video, holographic communication can convey the subtleties of non-verbal communication and provide a sense of presence and immediacy that enhances the quality of human interaction. More than 50 percent of smartphone users [2] expect this technology to be available to them within a few years.

Consumers are not the only ones longing for more authentic forms of digital communication. A recent study showed that the main barrier to remote working was the need for social interaction [3]. With the time spent working outside the office expected to increase in the coming decade, many office workers will require more immersive forms of digital interaction.

Starting with AR glasses and holographic calls with spatial audio, office workers around the world expect to benefit from the emergence of haptic technology, which adds tactile sensing of digital objects [3]. More than half of them say they would like a digital workstation with multisensory presence at the office when working remotely. Similarly, in a recent online survey of 7,115 self-defined early adopters aged 15-69 in 14 major cities, 80 percent of the respondents said they expect to have telepresence facilities to socialize with remote colleagues by 2030.

The potential for realizing holographic communication use cases in the years ahead is high. They are already anticipated in numerous consumer and enterprise areas ranging from attending family events as a hologram or meeting a doctor from home to remote presence at the office, expert help at a factory and immersive marketing. As soon as the ecosystem is ready to deliver these new experiences at a good price point and quality of experience (QoE), our research suggests that both private consumers and enterprises will be keen to use it.

The ability to make holographic communication an everyday reality is dependent on three key factors. Firstly, there needs to be a desire that drives the behavioral change (the human factor). Secondly, the appropriate AR devices need to become available. Lastly, mobile networks must have the ability to support the holographic communication pipeline.

The human factor

Users are the primary beneficiary of any type of human communication technology, and holographic communication is no exception. It is therefore essential to understand how they perceive it, along with the benefits from their perspective and their opinions about how to improve the user experience.

The two major user groups for holographic communication (enterprises and consumers) have different priorities. Enterprises will use holographic communication if it satisfies productivity goals better than existing tools. These can be defined as effectiveness, efficiency, and satisfaction in a specified context of use [4].

Consumers, on the other hand, tend to select the communication option that best satisfies their fun goals, characterized by hedonic, emotional and experiential perspectives [5]. Emotions are known to be an important guide to decision making [6]. It is therefore crucial that people's feelings are understood throughout the development process, which is achievable through user studies.

In an internal study on the topic, we discovered that a key emotion people feel about holographic communication is excitement. They are excited about seeing a holographic version of people they know in the same room as themselves, and they are excited about the future of this technology. Such a response is expected due to the novelty of the interaction, but as the novelty (or "halo") effect diminishes, people tend to prioritize factors such as usability, usefulness and familiarity [7].

> “A KEY EMOTION PEOPLE FEEL ABOUT HOLOGRAPHIC COMMUNICATION IS EXCITEMENT”

While some QoE metrics already exist for XR communication [8], several other tools are also available. Design thinking [9] can be used early on to ensure the best solution is built, while qualitative methods such as interviews can be used with early prototypes to provide deeper insights into how people feel. Other human-centered topics such as ethics, consent and accessibility should be considered and regularly monitored. Involving experts in human-computer interaction, ergonomics, psychology and user experience early in the design process will ensure that the final product is in line with the intended users' expectations.

It is essential that the human factor is prioritized in the development of holographic communication. One way to facilitate this would be to start referring to the end-to-end (E2E) pipeline as human to human rather than glass to glass.

Holographic communication

The area of holographic communication is divided into three paradigms: (1) avatars, (2) professional-quality digital representations of users and (3) consumer-friendly digital representations of users. "State-of-the-art avatar" technologies use artificial intelligence (AI) to generate a photorealistic representation of a user. Professional-quality digital representations are created through the use of multiple cameras and real-time studio recordings. Consumer-friendly digital representations are generated with the help of an AI-enabled, 3D-capturing setup on consumer grade phones or tablets. The third paradigm is the focus of this article.

	Use case	Typical usage (hours)	Resolution	Battery life	Mobility
Enterprise	Retail shopping	1-4	+++	++	+++
	Logistics (picking)	4-8	+	+++	+
	Augmented education	4-8	+++	+++	+
	Remote support	1-2	+++	++	+++
Consumer	AR gaming	1-2	+++	++	+++
	AR navigation	1-4	+	++	+++
	AR event (stadium)	1-2	+++	++	+++
Enterprise/ Consumer	Holographic communication	1-4	+++	+	+++

Figure 1 AR use cases and their requirements on the glasses ranked by importance (+ to +++)

Augmented reality glasses and other devices

Early AR glasses were connected by a cable, followed by the usage of Wi-Fi technology that limited mobility to home and office use in most cases. Over the last five years, however, the AR glasses market has been characterized by strong technology evolution with respect to all of the most important parameters: weight, field of view, resolution, battery life and mobility.

The first generation of lightweight AR glasses, which are connected through USB-C to 4G- and 5G-enabled smartphones, were launched two years ago. The next generation is expected to have built-in cellular connectivity, similar to today's smart watches. *Figure 1* highlights some of the enterprise and consumer use cases that will benefit most from next-generation AR glasses.

There are two different AR device types: those with SLAM (simultaneous localization and mapping) support and those without it. SLAM uses one or more front cameras to create and update a map of the surroundings that allows the device to localize itself and to anchor virtual content (including holograms) on fixed locations in the environment. A true holographic impression is generated by the ability to see the hologram from different angles, which can be achieved with SLAM.

The ongoing evolution of both mobile and stationary holographic displays as well as AR-enabled contact lenses will also contribute to the uptake of holographic communication. AR-enabled tablets and phones allow a lower barrier to entry in cases where hands-free operation is not required.

Terms and abbreviations

AI – Artificial Intelligence | **AR** – Augmented Reality | **E2E**– End-to-End | **FPS** – Frames Per Second | **IEC** – International Electrotechnical Commission | **ISO** – International Organization for Standardization | **LiDAR** – Light Detection and Ranging | **MPEG** – Moving Picture Experts Group | **MS** – Millisecond | **NR** – New Radio | **QoE** – Quality of Experience | **RAN** – Radio Access Network | **SLAM** – Simultaneous Localization and Mapping | **ToF** – Time of Flight | **V3C** – Visual Volumetric Video-based Coding | **V-PCC** – Video-based Point Cloud Compression | **XR** – Extended Reality

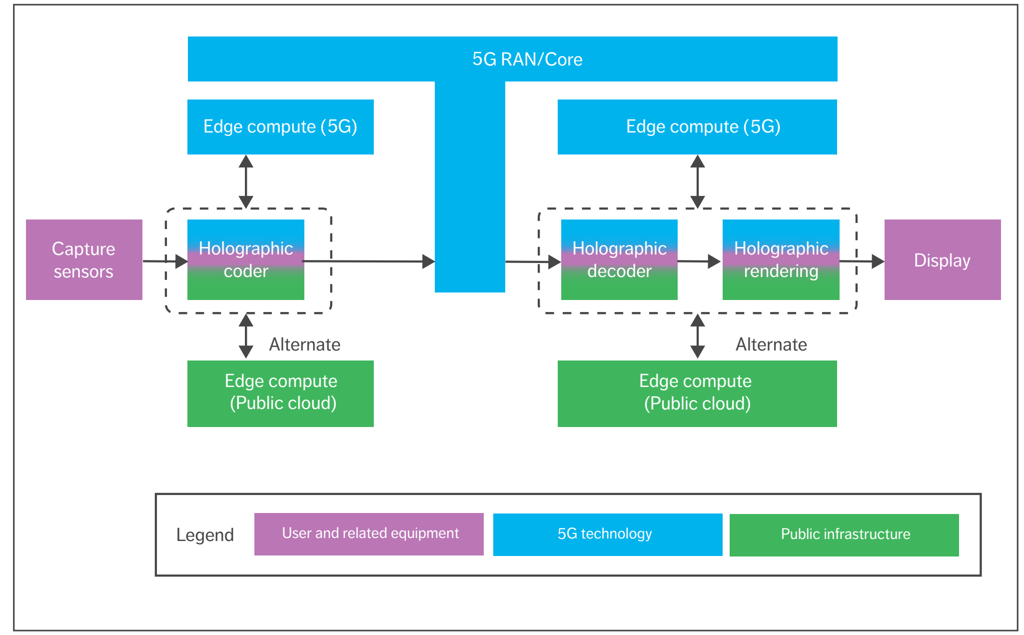

Figure 2 End-to-end pipeline for holographic communication

Holographic communication pipeline

Our proposed architecture to support holographic communication is depicted in *Figure 2*. In it, capturing sensors provide a real-time representation of the human face and body. Format conversion and filtering is applied before encoding to reduce the bitrate requirements on the network. The compressed hologram is transmitted to an XR device over a low latency reliable transport network such as 5G. On the XR device, the compressed hologram is decoded and processed before rendering in the consumer environment. The rendering engine considers positioning and semantic information of the device and the rendered scene. The virtual human representation is displayed on the XR device. The network can act as a compute platform for the encoding, decoding and rendering of holographic data to offload heavy processing from the devices.

Capturing technologies

Holographic capturing is the process of creating a measurable 3D representation of an object, person or environment. This process is divided into four steps:

1. Acquisition
2. Depth estimation
3. Data fusion
4. Post-processing.

The acquisition step makes use of visual sensors to capture a volume of interest. Several different visual sensor technologies are used for 3D capturing. Nowadays, the most common mechanism is the usage of time-of-flight (ToF) sensors (such as

MESHES CAN BE SIGNIFICANTLY SMALLER THAN POINT CLOUDS, INCREASING THE SPEED OF STORAGE, TRANSMISSION AND RENDERING

LiDAR), which measure distance by the time it takes a light pulse to travel to its destination and back.

In the depth estimation step, sensor streams are used to compute depth. ToF sensors can provide depth information directly, while stereo cameras and multi-camera systems estimate depth by capturing the person from different angles.

In the data fusion step, the depth information or depth map from different perspectives is fused into a single stream of 3D points by matching key points and calculating the optimal geometric transformation from the different views.

The post-processing step reduces the data size of the stream of 3D points by cleaning up redundant points, noise and outliers. The resulting 3D representation can be delivered in various visual media formats such as point clouds or meshes.

A point cloud is a collection of points that represent the captured volume. Each point contains information on its location, as well as the option of the red, green and blue color model and intensity value at a specific frame. A mesh unites those points with triangles, disregarding redundant points and filling any holes. Meshes can be cleaned and reduced further by decreasing the number of vertices. Depending on their resolution, they can be significantly smaller than point clouds, increasing the speed of storage, transmission and rendering of meshes in comparison with point clouds.

Rendering and display

Rendering is the process of generating an image of a scene or a model from a given viewpoint using computations. A scene is a container object describing a volume and its contents. Sources are objects located in the scene to be rendered. A camera is an instance that renders the viewpoint and consists of a location, focus, orientation and resolution.

Engines render the content based on rendering pipelines. The rendering pipelines take care of culling, rendering and post-processing. There are several types of pipelines with different capabilities and performance characteristics that are suitable for different applications and platforms. A general-purpose render pipeline is optimized to process graphics across a wide range of platforms, while a high-fidelity graphics pipeline is suitable for high-end platforms.

Additional techniques and approaches are available to improve the rendering process in various ways, such as enabling faster rendering with better QoE, smoothing squared edges or increasing the quality of objects in a scene. Furthermore, AI algorithms can recreate an object of a scene or create a photorealistic representation [10] such as an avatar. The photorealistic representation contains a model: a visual representation or mesh model and configurations. Configurations are the bones or access points of a model. These points are animated during the rendering process with preloaded animations.

Real-time data captured from a depth camera takes more computation for rendering compared with an avatar representation. The rendering of all parts of the mesh is needed for every frame for a hologram, compared with rendering a difference for an avatar, such as updating facial expressions like the blinking of an eye on an existing avatar model.

Split rendering is an approach that offloads rendering capabilities to an edge cloud. The key idea is that the six degrees of freedom position and orientation of the XR device in a scene are tracked, and constant feedback is provided to the edge in real time. The rendering of a 3D scene happens in a cloud. Based on the position of a user, the cloud streams back a 2D video to a scene from the user's angle. For this approach, an end-user device does not need to have high-end characteristics. However, good QoE in this scenario requires low latency communication between the edge and the device.

REAL-TIME DATA CAPTURED FROM A DEPTH CAMERA REQUIRES MORE COMPUTATION FOR RENDERING

When it is ready, the rendered stream is transmitted to a device with the ability to provide users with a holographic experience. There are four device types: handheld devices (such as smartphones and tablets), holographic displays, AR glasses and virtual reality glasses. Using these devices, it is possible to place a hologram in a room and move around it.

Media formats and codecs

The delivery of holographic communication will require the processing and transfer of various visual media formats. Those formats aim to represent more realistic and interactive visual representations of humans and/or an environment than established 2D video formats used in traditional videoconferencing, yet the increased informational load may put high pressure on transmission bitrates across the whole communication chain.

For example, point cloud data representing a single human can typically consist of 100,000 to 1 million points and beyond per single time instance (video frame). Streaming such data with 30 frames per second (fps) – the typical videoconferencing streaming rate – would require roughly between 300Mbps to 3Gbps of available bandwidth. Such uncompressed bitrates are not feasible today, and current real-life videoconferencing systems employ media codecs to reduce the bandwidth requirements to single Mbps (1-6Mbps) offering compression ratios from 250:1 up to 1000:1.

Methods to offer similar compression ratios for immersive 3D representations such as point clouds are therefore needed to enable the deployment of holographic communication services. In terms of handling 3D visual formats and their decoding on XR devices, two scenarios can be considered.

IT IS POSSIBLE TO PLACE A HOLOGRAM IN A ROOM AND MOVE AROUND IT

In the first, which is typically applied in the case of split rendering, the processing and decoding of 3D formats is done at the edge, while XR devices decode pre-rendered 2D video with traditional 2D video codecs such as ITU-T (International Telecommunication Union Telecommunication Standardization Sector) and ISO/IEC (International Organization for Standardization/International Electrotechnical Commission) MPEG (Moving Picture Experts Group) High Efficiency Video Coding/H.265 or Versatile Video Coding/H.266 codecs.

In the second scenario, the processing and decoding of 3D formats is done on XR devices requiring the support of additional immersive decoders on the device itself. One approach standardized in ISO/IEC MPEG-I visual volumetric video-based coding (V3C) and video-based point cloud compression (V-PCC) employs 3D to 2D projection algorithms that create intermediate 2D video representations. These representations can be decoded with multiple instances of 2D video codecs such as H.265 or H.266.

An alternative approach would be to employ "native" 3D codecs such as the ISO/IEC MPEG-I geometry-based point cloud compression. In this codec, compression tools operate directly on a 3D point cloud representation. Unlike for the V3C/V-PCC codec, this approach would require adding support for such native point cloud codec(s) in mobile hardware chipsets.

Test results

To determine the feasibility of using existing 5G technology to build an E2E pipeline that enables high-quality holographic communication, we tested the approach presented in Figure 2 in a live 5G New Radio (NR) network.

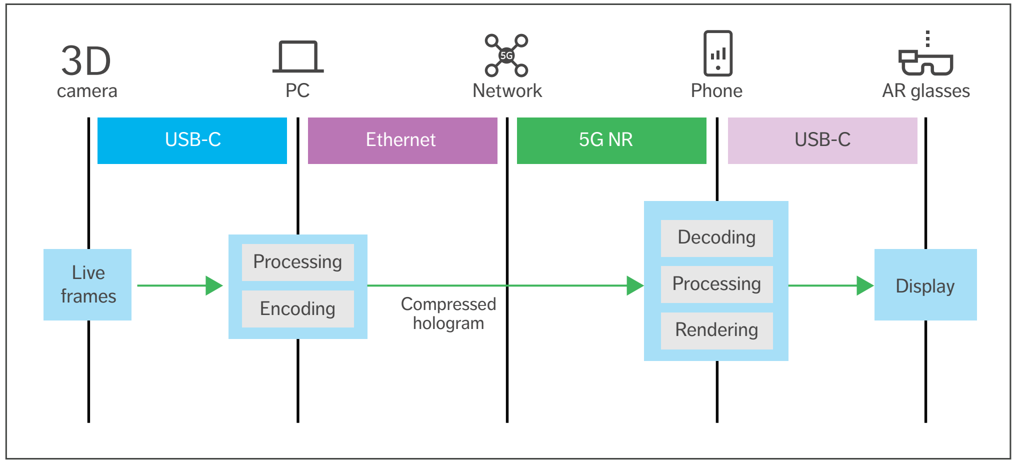

Figure 3 The test setup in the lab

The 3D frames were captured by a single 3D camera connected to a computer, which was connected by ethernet to the 5G network. The captured stream was compressed and sent over the 5G network to a 5G mobile phone with connected AR glasses. The decoding and rendering were executed on the mobile phone and displayed in the AR glasses. The SLAM functionality made it possible to walk around the 3D representation. The measurement setup is shown in *Figure 3*.

Factors that affect QoE in holographic communications include bandwidth, latency and quality. These three are collectively referred to as the trade-off triangle because only two of them can be optimized at the same time, and it will always be at the cost of the third. In our measurements, we chose to change the quality and bandwidth.

> FACTORS THAT AFFECT QOE IN HOLOGRAPHIC COMMUNICATIONS INCLUDE BANDWIDTH, LATENCY AND QUALITY

The quality was changed by adjusting the number of points in the point cloud (10,000 points/30fps to 1 million points/30fps). The compression rate was kept constant resulting in bandwidth of between 5Mbps and 100Mbps.

The total latency in the phone is determined by two factors: the decoding and the rendering. As expected, the latency increases when the quality and bandwidth increase. It is clearly visible that the compute in the mobile phone is reaching its limit for the high quality, causing a latency of approximately 170ms (milliseconds). The first experiments show that a move of the decoding compute from the phone to the edge cloud can reduce the latency to 70ms.

Figure 4 shows the bitrate, round-trip time and phone latency results for four quality levels (Q1-Q4) all with 1024x780 resolution and 30fps. The average number of points for each was Q1 (900,000), Q2 (225,000), Q3 (100,000) and Q4 (36,000).

Conclusion

Our research indicates that both consumers and enterprises understand the potential of immersive, augmented reality (AR) applications, and holographic communication in particular, to improve their lives not only in terms of workplace productivity, but also when it comes to social interaction and entertainment. The emergence of lightweight AR glasses and powerful 3D compression algorithms have made it possible to start deploying AR use cases using existing 5G technology.

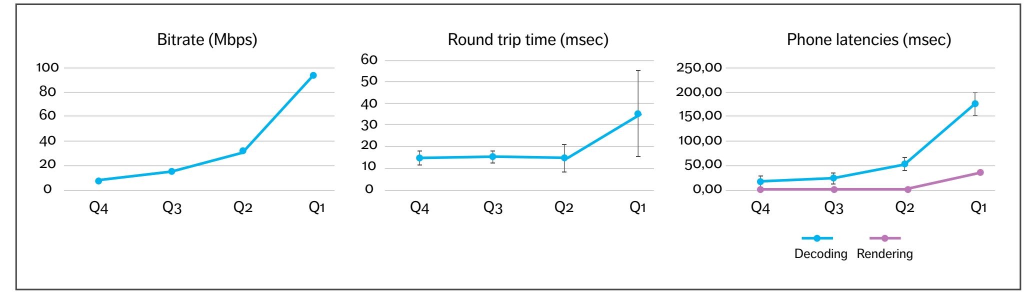

Figure 4 5G NR measurement results

An architecture with edge compute functionality across the full chain is necessary to deliver the 3D captured streams and the rendering of such applications anywhere, while simultaneously meeting device requirements on size, weight and energy consumption. This approach makes it possible to move high performance computing to the network and thereby reduce both the energy consumption of mobile devices and the end-to-end latency.

The introduction of new media formats such as point clouds into extended reality (XR) communication scenarios will greatly enhance the attractiveness, usefulness and efficiency of the information conveyed between communicating parties. Successful communication requires a joint understanding of the communication format, which is typically solved through standardization. In Release 18, the 3GPP has taken on the challenging task of enhancing 5G to offer more efficient support for XR services. Since various aspects of XR are already being addressed in other standardization forums, the intention is to reuse as much as possible.

Further reading

- **Ericsson Technology Review, XR and 5G: Extended reality at scale with time-critical communication, available at:** *https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/xr-and-5g-extended-reality-at-scale-with-time-critical-communication*
- **Ericsson Technology Review, Versatile video coding explained, available at:** *https://www.ericsson.com/en/reports-and-papers/ericsson-technology-review/articles/versatile-video-coding-explained*
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EXTENDED REALITY AND 5G ✱

XR and 5G: Extended reality at scale with time-critical communication

The value of 5G extends far beyond the enhanced mobile broadband that more than 1 billion people already have access to through upwards of 150 communication service providers around the world [1]. The time-critical communication capabilities in 5G networks will enable major breakthroughs in a wide range of application areas, including extended reality (XR).

FREDRIK ALRIKSSON, DU HO KANG, CHRIS PHILLIPS, JOSE LUIS PRADAS, ALI ZAIDI

In contrast to enhanced mobile broadband (MBB), the majority of the emerging 5G value-adding applications are time critical in nature with demanding requirements on reliable low latency. Time-critical use cases can be broken down into four categories: real-time media, remote control, industrial control and mobility automation [1, 2].

■ The real-time media category includes innovative extended reality (XR) applications that are of growing interest for consumers, enterprises and public institutions alike. The emergence of time-critical communications over 5G networks makes it possible to offload parts of the XR processing and functionality to the edge cloud and enhance the user experience with lightweight and cost-efficient head-mounted displays (HMDs).

XR use cases

XR is an umbrella term that covers immersive technologies ranging from virtual reality (VR) to mixed reality (MR) and augmented reality (AR). In VR, users are totally immersed in a simulated digital environment or a digital replica of reality. MR includes all variants where virtual and real environments are mixed. AR is one such variant, where digital information is overlaid on images of reality viewed through a device. The level of augmentation can vary from a simple information display to the addition of virtual objects and even complete augmentation of the real world. MR can also include variants where real objects are included in the virtual world.

XR is expected to improve productivity and convenience for consumers, enterprises and public institutions in a wide variety of application areas such as entertainment, training, education, remote support, remote control, communications and virtual meetings. It can be used in virtually all industry segments, including health care, real estate, shopping, transportation and manufacturing. VR is already used for gaming both at home and at dedicated venues, for virtual tours in the context of real estate, for education and training purposes and for remote participation at live events such as concerts and sports.

While VR holds great promise, AR and MR use cases have even greater transformational potential. In VR, the headsets cut users off from their physical surroundings and restrict mobility [3]. With AR, users are present in reality and free to move even when using HMDs. Many smartphone users have already experienced basic forms of AR, through games like Pokémon Go and apps that enable shoppers to visualize new furniture in their homes before making a purchase. AR technology becomes much more powerful, though, when it is used with HMDs. By freeing up the user's hands, AR HMDs transform the user interaction. The ability to have information overlaid on the real world while simultaneously having your hands free has been shown to increase worker efficiency dramatically [4].

> “WHILE VR HOLDS GREAT PROMISE, AR AND MR USE CASES HAVE EVEN GREATER TRANSFORMATIONAL POTENTIAL”

XR edge-processing architectures

VR HMDs are already available at scale on the market today, but they are rapidly evolving. Because VR applications are often processing-intensive, enabling high-end VR requires connecting the HMDs to high-end processing units – typically powerful PCs or gaming consoles. VR HMDs with local processing capabilities are also starting to become available, but these are relatively large and heavy and cannot provide the same experience as when off-device processing is used.

AR HMDs are also available on the market today, predominantly for enterprise use [5]. Mass-market adoption will require further progress on ease of use, attractive appearance and content availability [6]. Building fashionable, small form factor HMDs to meet the demands for XR is challenging due to limited processing power, storage, battery life and heat dissipation. We believe that the best way to address these challenges is by offloading parts of XR processing to the mobile network edge.

Terms and abbreviations

5GC – 5G Core | **AAS** – Advanced Antenna System | **AR** – Augmented Reality | **CG** – Configured Grant | **CoMP** – Coordinated Multi-Point | **DAPS** – Dual Active Protocol Stack | **DC** – Data Center | **DL** – Downlink | **E2E** – End-to-End | **L4S** – Low Latency, Low Loss, Scalable Throughput | **HMD** – Head-Mounted Display | **MBB** – Mobile Broadband | **MR** – Mixed Reality | **SPS** – Semi-Persistent Scheduling | **TRP** – Transmission Reception Point | **TTI** – Time Transmission Interval | **UE** – User Equipment | **UL** – Uplink | **VR** – Virtual Reality | **XR** – Extended Reality

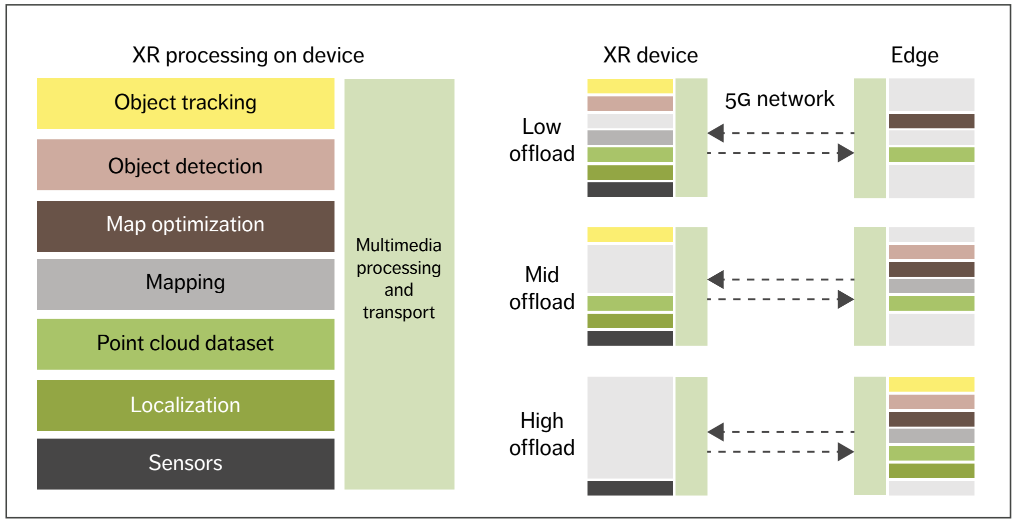

Figure 1 Split architecture options with 5G connectivity

Figure 1 shows the eight main types of XR functionality and how they can be split up between an XR device and the network edge. The major components in XR processing include SLAM (simultaneous localization, mapping and map optimization) [7] with point cloud datasets [8], hand gesture and pose estimation [9], object detection and tracking [10] and multimedia processing and transport. Examples of multimedia processing and transport are rendering, asynchronous time warp [11] and video, audio and sensor encoding.

Figure 1 also illustrates three architectures for splitting the XR processing: low offload, mid offload and high offload. Our internal studies indicate that low offload reduces device energy consumption by threefold while mid offload reduces it by fourfold. High offload reduces device energy consumption by more than sevenfold.

OUR INTERNAL STUDIES INDICATE THAT HIGH OFFLOAD REDUCES DEVICE ENERGY CONSUMPTION BY MORE THAN SEVENFOLD

In the low-offload architecture, almost all processing is done on the device. The point cloud dataset, spatial map generation and localization occur on the device. As portions of the point cloud and spatial map are built, the point cloud is compressed through multimedia processing and transmitted over the 5G network to the edge, where uploaded spatial map data is created and merged into existing global spatial map data. The object detection and tracking are performed on the device. The rendering can occur locally on the device or at the network edge.

In the mid-offload case, the localization and object tracking functions are performed on the device. The spatial map and point cloud datasets are generated in the network. Key image frames are compressed using a video codec and sent to the network edge to generate the spatial map and point cloud datasets, and to perform object detection. At the network edge, the point cloud datasets and spatial map are created and merged with the global point cloud

THE XR CONNECTIVITY REQUIREMENTS DEPEND ON THE LEVEL OF SPLIT ARCHITECTURE AND THE TARGETED QoE

datasets and spatial map. The overlay rendering occurs at the network edge. The edge-rendered video and audio content is encoded into video and audio streams and transmitted to the device along with the rendering mesh data.

In the high-offload case, only sensor data is sent over the uplink. Sensor data includes camera data. Many AR/VR devices have multiple cameras, including infrared and RGB (red, green and blue) ones. Sensor data also covers sensors such as LiDAR (light detection and ranging) and IMU (inertial measurement unit). The image data is encoded using video compression such as Motion Pictures Experts Group (MPEG) High Efficiency Video Coding (HEVC) or Versatile Video Coding (VVC).

Several compression techniques have been proposed for IMU data, such as delta encoding, linear extrapolation, second- to fifth-order polynomial regression and spline extrapolation [12]. There are numerous techniques for compressing 3D point cloud generated by LiDAR sensors. ISO/MPEG currently has two tracks for point cloud compression standardization under development. These are Video-based Point Cloud Compression (V-PCC) and Geometry Based Point Cloud Compression (G-PCC). LiDAR is covered in the MPEG G-PCC track [13]. The edge-rendered video and audio content is encoded into video and audio streams and transmitted to the device along with the rendering mesh data.

XR traffic characteristics and connectivity requirements

XR traffic is characterized by a mixture of pose and video from/to the same XR device, varying video frame size over time and quasi-periodic packet arrival with application jitter after IP segmentation. Traffic arrival time to the RAN is periodic with non-negligible jitter due to application-processing-time uncertainty. Video frame sizes are an order of magnitude larger and, at the same time, not fixed over time compared with packets in voice or industrial control communication. The segmentation of each frame is expected, which implies that packets arrive in bursts that must be handled together to meet stringent bounded latency requirements.

XR connectivity requirements depend on the level of split architecture and the targeted QoE, leading to a wide range of bit rates and bounded latency requirements. *Figure 2* presents the 5G connectivity requirements for AR, VR and cloud gaming based on the developments in the ecosystem, including the 3GPP [14]. The requirements assume local processing techniques in the split architecture to

Use cases	DL bitrates (Mbps)	UL bitrates (Mbps)	One-way latency (ms)	Frame realibility (%)
Cloud gaming	8-30	~0.3	10-30	≥99
VR	30-100	< 2	5-20	≥99
AR	2-60	2-20	5-50	≥99

Figure 2 Use-case requirements for 5G networks

mitigate consumer latency requirements [15]. Note that latency and reliability requirements are on a video frame (or file) level excluding application error and delay.

For downlink (DL) video traffic, VR typically needs higher bit rates than cloud gaming to support retinal resolution when using HMD and lower compression efficiency due to low-latency encoding. Some AR applications for conversational services can have DL video traffic as VR. However, they potentially have lower resolution to render a video on only part of display, leading to a lower bit rate than VR DL video. In addition, they can have uplink (UL) video streaming for the object detection and tracking, but the bit rate requirements can be lower compared with the DL video. All cloud gaming, AR and VR include pose traffic in the UL, which has much lower bit rates than video traffic, but AR and VR will have higher bit rate requirements than cloud gaming to convey more pose information, such as six degrees of freedom.

AR and VR require more stringent end-to-end (E2E) bounded latencies than cloud gaming since a human is more sensitive to the discrepancy in 3D virtual environments. For instance, it is widely accepted that rendering motion to photon latency greater than 20ms starts to cause nausea when a human wears a VR HMD. Processing techniques such as asynchronous time warp [11] relax the latency requirement to some extent, making VR feasible over 5G networks. For AR, object detection can be performed at the network edge, and object tracking can be done on the device as shown in the low- and mid-offload cases in Figure 1. By performing object detection in the network and tracking on the device, the bounded latency requirement for a 5G network can be relaxed up to 50ms.

Network architecture for time-critical communications

Time-critical communications is an emerging 5G concept for enabling services with reliable low latency requirements such as XR [2]. The aim is to secure data delivery within specific latency bounds (X ms) with the desired reliability level (Y percent). Depending on the user requirements, X ranges from tens of milliseconds to 1 millisecond and Y ranges from 99 percent to 99.999 percent reliability. To ensure bounded latencies, the system may have to compromise on capacity, throughput, energy efficiency or coverage.

The 5G RAN, the 5G Core (5GC) and the transport network together with the device contribute to the E2E reliability and latency. E2E latency is the sum of individual latency contributions from every component. E2E reliability cannot be better than the reliability of the weakest link.

Edge deployment of 5GC and applications is key to reducing the transport latency between the application and the RAN. If an application is hosted in a central national data center (DC), the transport network round-trip latency can be in the order of 10-40ms, depending on the distance to the DC and how well the transport network is built out. The transport latency can be reduced to 5-20ms by moving applications to a regional DC or even to 1-5ms for edge sites. For local network deployments with networking functions and applications hosted on-premises, transport latencies become negligible [2].

Achievable RAN latency/reliability performance depends on general deployment factors (such as frequency band, bandwidth, inter-site distances, numerology, duplexing schemes and TDD configuration), RAN and user equipment (UE) capabilities (in terms of hardware and software features) and traffic characteristics (such as data rate and packet size).

5G toolbox to achieve time-critical communications

To ensure that XR applications work well over 5G networks, it is important to separate the XR traffic from best-effort MBB traffic using the 5G QoS framework with optimized QoS flows, as illustrated at the top of *Figure 3*. This enables optimized treatment of XR throughout the mobile network and specifically in the RAN to mitigate the different sources of delay.

The provision of bounded latency for XR and

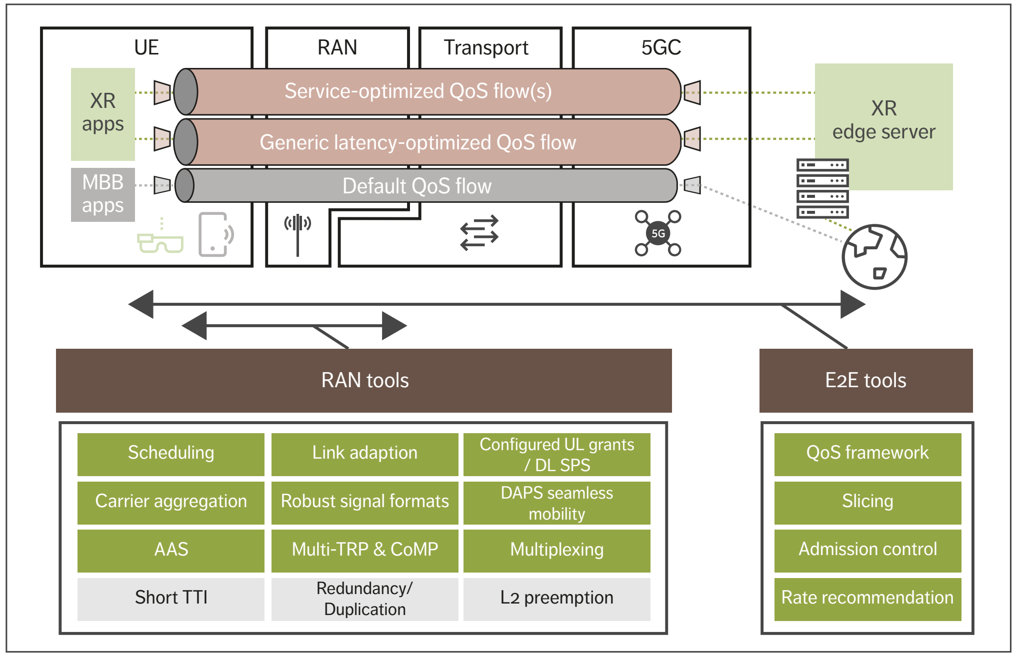

Figure 3 5G toolbox to realize time-critical communications – the tools that are important for XR are marked in green

other services requires the prevention of delays and interruptions. The bottom half of Figure 3 presents the comprehensive 5G toolbox for realizing time-critical communications that makes it possible to overcome the following five main causes of delays and interruptions:

1. Congestion
2. Dynamic radio environment
3. Standards/protocols
4. Mobility
5. Device power saving.

Congestion

There are several ways to mitigate the congestion-related delays that can occur when end hosts transmit at a higher bitrate than the network can sustain. Fast application rate adaptation is key to avoid congestion for rate-adaptive traffic such as XR and cloud gaming. Low Latency, Low Loss, Scalable throughput (L4S) is an existing method that provides fast indication of congestion from networks that can be applied in the RAN, enabling RAN rate recommendation and forming the basis for a common framework for rate adaptation over 5G [16]. Network slicing and radio resource partitioning together with admission control and latency-optimized scheduling are important tools for reserving network resources for time-critical services and avoiding congestion-related delays to provide a minimum guaranteed bit rate, for example. Network slicing is also beneficial for protecting MBB and other services from resource-hungry, time-critical services.

Dynamic radio environment

Advanced scheduling together with robust link adaptation and signal transmission formats are important tools to combat delays related to the

EXTENDED REALITY AND 5G ✱

THE 5G QoS FRAMEWORK MAKES IT POSSIBLE TO ESTABLISH QoS FLOWS THAT PROVIDE OPTIMIZED NETWORK TREATMENT FOR SPECIFIC FLOWS

dynamic radio environment such as fading/blocking and interference. High-rate XR traffic places new demands on these tools to deliver very high spectral efficiency, while ensuring reliable and timely video frame delivery. Advanced antenna systems (AASs) have tremendous potential to improve the link budget, reduce interference, and increase spatial multiplexing, ultimately leading to more radio system capacity for XR.

Standards/protocols

Features that minimize delays associated with standards/protocols include tools like prescheduling, UL configured grants (CGs) and DL semi-persistent scheduling (SPS). As an example, a combination of periodic CGs and dynamic grants can be used to reduce latency for UL AR traffic consisting of periodic frames of varying size. XR traffic arrival time characteristics call for further optimizations of UL CGs and DL SPS to avoid unnecessary waiting times. Protocol enhancements throughout the different protocol layers, from the SDAP (Service Data Adaptation Protocol) to the physical layer, including the PDCP (Packet Data Convergence Protocol), the RLC (Radio Link Control) and the MAC (Medium Access Control) protocols, can be important for improving the capacity of XR applications as well. For example, control signaling efficiency could be improved to provide grants for multiple radio resource allocations needed for a large XR video frame.

Mobility

Time-critical services place much more stringent requirements on mobility performance than MBB services. For XR services, mobility interruptions need to be well below the inter-frame arrival time (which is typically between 15-50ms) to pass unnoticed. This will require smarter and faster network algorithms as well as stricter processing requirements at the device. 5G New Radio provides a few options for supporting seamless and more robust mobility, including multiple transmission reception points (multi-TRPs), dual active protocol stack (DAPS) handover and conditional handover.

Device power saving

Device power saving is important to support low-power XR devices, and XR traffic arrival time characteristics also call for further optimizations of discontinuous reception.

Traffic awareness in the RAN

One interesting area of future standardization work in the 3GPP is to further optimize 5G performance and capacity by improving XR traffic awareness in the RAN, especially for very short-term traffic variation that may be difficult for an application layer to handle. If, for example, the RAN had the ability to know which IP packets are associated with the same application frame, it could potentially use that knowledge to optimize radio resource allocation, scheduling, link adaptation, packet discarding and other features to increase capacity.

5G QoS approaches for extended reality

The 5G QoS framework makes it possible to establish QoS flows that provide optimized network treatment for specific traffic flows, in addition to the default QoS flow used for MBB. Such additional QoS flows can be established either using 5GC QoS-exposure application programming interfaces to communicate service requirements, or by traffic detection together with pre-provisioned service requirements, such as relying on standardized 5G QoS identifier characteristics. Two complementary 5G QoS approaches can be implemented for XR: a generic latency-optimized QoS flow and service-optimized QoS flows.

RAN deployment	Frequency allocation	Min. bit rate	Max. latency	Frame reliability
Wide area	Mid-band FDD 2x20MHz@2GHz Mid-band TDD 100MHz@3.5GHz	DL: 8-30Mbps UL: 2-10Mbps	DL: 10-30ms UL: 10-30ms	99%
Indoor	Mid-band TDD 100MHz@3.5GHz	DL: 30-60Mbps UL: 10-20Mbps	DL: 10-30ms UL: 10-30ms	99%
	mmWave 800MHz@30GHz		DL: 5-10ms UL: 5-10ms	

Figure 4 Simulation assumptions for 5G RAN

Generic latency-optimized QoS flow

A generic latency-optimized QoS flow that can be implemented in any network is an important tool to enable a large ecosystem of high-rate, rate-adaptive, time-critical applications and services including XR to emerge and hopefully flourish in the same way as MBB and smartphones have done [16]. Implementing such a generic QoS flow enables networks to provide L4S-based early RAN congestion detection and fast rate recommendation to applications as well as packet treatment optimized for low latency and jitter rather than throughput. By not relying on knowledge of specific service requirements it avoids a tight coupling between application and network.

Service-optimized QoS flows

Service-optimized QoS flows can be established for specific XR services with known service requirements in terms of packet delay budgets, packet error rates, minimum guaranteed bit rates and so on, in cases where there is a need to increase coverage or capacity while still providing a good minimum QoE, for example. This requires a tighter coupling between application and network, which adds complexity to the ecosystem but provides QoE beyond what the generic latency-optimized QoS flow can enable and may be required for the most demanding XR applications.

Deployment strategy

To develop insights regarding the 5G network deployment strategy for various XR applications, we have carried out simulation studies for a wide-area deployment and an indoor enterprise deployment. For the wide-area scenario, we evaluated capacity for low- to mid-range XR applications in terms of requirements. For the indoor deployment, we considered high-end applications as well. The simulation assumptions are summarized in *Figure 4*. The 99 percent frame reliability implies less than 0.1 percent packet error rate if each frame is segmented into more than 10 IP packets.

The wide-area scenario is based on a macro deployment in central London with an inter-site distance of approximately 450m. For the mid-band, wide-area deployments, we include an AAS with 32 elements and 16 elements per polarization for 3.5GHz and 2GHz, respectively. The indoor deployment also assumes eight elements and 32 elements per polarization of an AAS for 3.5GHz and 30GHz, respectively. Devices with four receiver

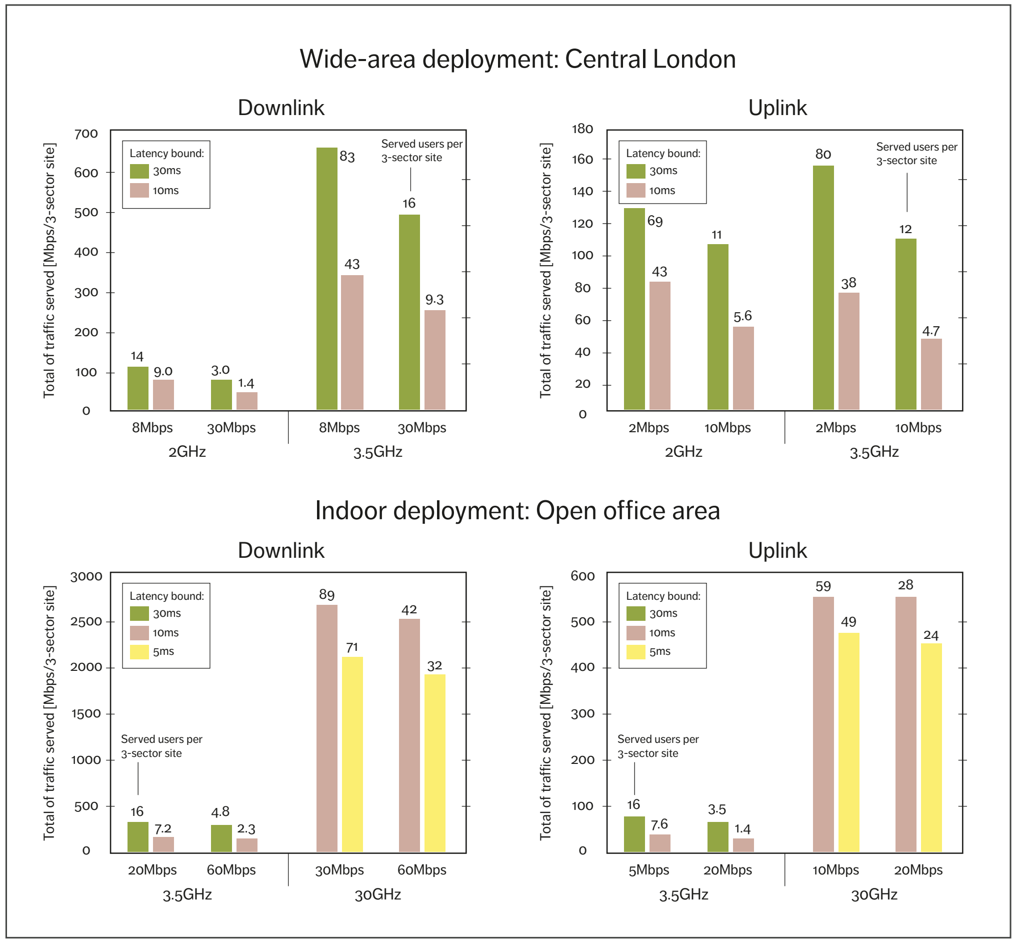

Figure 5 5G RAN-served traffic for various requirements

branches were used in the evaluation. The indoor deployment assumes a 120m by 50m open office area, where four pico sites with three sectors each are placed on the ceiling. For the TDD bands, we have assumed 4:1 DL and UL configuration.

Figure 5 shows the average served traffic per cell (with different combinations of the minimum bit rate and maximum latency requirements per user) for the wide-area and the indoor deployment scenarios assuming 90 percent and 95 percent coverage availability, respectively. From the cell throughput and the minimum bit rate per user, it is possible to derive the maximum number of users served simultaneously per cell, which is shown next to each bar in Figure 5. When the cell capacity is underutilized, the applications can increase the bit rate for superior QoE by rate adaptation.

We can observe that the cell capacity in the traffic served decreases by increasing the minimum user bit rate requirements or by reducing the maximum latency target. Increasing the user bit rate requirement at a given latency budget creates more interference and resource utilization, leading to a smaller amount of total traffic being served. Similarly, when reducing the latency requirement for a given bit rate requirement, a more stringent latency target implies that the resources are available on a shorter time scale, which leads to fewer users being served.

These results show that communication service providers can start to address cloud gaming and low-end AR use cases (for consumers and enterprises) in a wide area with the ongoing 5G network rollout utilizing mid bands, and gradually evolve capabilities and densify deployments to address greater coverage and more demanding requirements in terms of throughput and bounded latency. For users with coverage needs in a small geographic area such as a theme park, industry campus or office, there is an opportunity to support high-end XR with indoor deployments and address more stringent latency and higher bit rate requirements with a local breakout of the core network.

Conclusion

Although extended reality (XR) has the potential to be transformational for both business and society, widespread adoption has previously been hindered by issues such as heat generation and the limited processing power, storage and battery life of small form factor head-mounted devices. The time-critical communication capabilities in 5G make it possible to overcome these challenges by offloading XR processing to the mobile network edge. By capitalizing on their ongoing 5G rollouts, mobile network operators are in an excellent position to enable the realization of XR on a large scale.

Further reading

- **Switch on a better 5G network, available at:** *https://www.ericsson.com/en/5g/5g-networks*
- **5G by Ericsson, available at:** *https://www.ericsson.com/en/5g*
- **How 5G and Edge Computing can enhance virtual reality, available at:** *https://www.ericsson.com/en/blog/2020/4/how-5g-and-edge-computing-can-enhance-virtual-reality*
- **A technical overview of time-critical communication with 5G NR, available at:** *https://www.ericsson.com/en/blog/2021/2/time-critical-communication--5g-nr*

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